End-to-End Fine-Grained Arabic Dialect Classification

A Comparative Study of Classical Stacking Ensembles and Pre-Trained Transformer Architectures

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Project Repository: github.com/meedooatef22/arabic-dialect-classification

Target Dialects: Beirut (Levantine), Cairo (Egyptian), Doha (Gulf), Tunis (Maghrebi), and Modern Standard Arabic (MSA)

🏆 **Champion Model:** `Fine-Tuned MARBERTv2` (**92.25% Unseen Test Accuracy** | **92.24% Macro F1-Score**)

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1. Executive Summary & Abstract

The Arabic language exhibits a complex sociolinguistic phenomenon known as **Diglossia**, where a standardized formal variety—Modern Standard Arabic (MSA)—coexists alongside numerous regional spoken dialects (*Ammiya*). With the rapid proliferation of social media and conversational AI platforms across the Arab world, speakers predominantly communicate in regional dialects that lack standardized orthography, exhibit heavy morphological agglutination, and share substantial lexical overlap. Consequently, standard Natural Language Processing (NLP) systems trained exclusively on MSA experience severe performance degradation when deployed on real-world user-generated content.

This graduation project designs, implements, and deploys a complete end-to-end NLP pipeline for fine-grained Arabic Dialect Identification (ADI) across five major varieties derived from the **MADAR Corpus**: **Beirut**, **Cairo**, **Doha**, **MSA**, and **Tunis**. We establish a rigorous zero-leakage experimental framework (**70% Train / 15% Validation / 15% Unseen Test**) and conduct a systematic benchmark across three modeling paradigms:

1. **Classical Machine Learning Baselines & Stacking Ensemble** using a hybrid Word `(1,3)` and Character-Boundary `(2,5)` TF-IDF representation (`150,000` features), achieving **90.93% Test Accuracy**.
2. **AraBERTv02 (`aubmindlab/bert-base-arabertv02`)** fine-tuning, achieving **91.03% Validation Accuracy**.
3. **MARBERTv2 (`UBC-NLP/MARBERTv2`)** fine-tuning, which establishes the project benchmark at **92.25% Test Accuracy** and **92.24% Macro F1-Score**.

Finally, we perform a deep quantitative and qualitative error analysis on the misclassified samples to prove the theoretical accuracy ceiling of short conversational text, and deploy our models into a production-ready **Streamlit Web Application** (`streamlit_app.py` & `app.py`) supporting real-time single-sentence analysis, head-to-head model comparison, and batch CSV inference.

2. Problem Statement & Linguistic Challenges

Classifying Arabic dialects at the sentence level presents four primary computational challenges:

1. **Lack of Orthographic Standardization:** Dialectal words have no single dictionary spelling; a single word may be written with different hamza forms (ا / آ / إ / أ), terminal vowels (ي / ى), or ta-marbuta variations (ه / ة) depending on the writer.
2. **Morphological Richness & Clitic Agglutination:** Dialectal Arabic attaches negation particles, prepositions, and pronouns directly onto stems (e.g., Tunisian فماش, Levantine شورايك, Gulf هالفيش), creating high out-of-vocabulary (OOV) rates for word-level models.
3. **Lexical Overlap & Polysemy:** Neighboring dialects (particularly Levantine `Beirut` and Gulf `Doha`) share identical vocabulary items that differ only in subtle contextual usage or pronunciation not captured in unvocalized text.
4. **Short Conversational Utterances:** Travel and social media sentences are frequently ultra-short (2 to 4 words), offering minimal contextual clues for disambiguation.

3. Dataset Description & Exploratory Data Analysis (EDA)

3.1 Dataset Overview

The project utilizes a balanced 5-class corpus consisting of **40,000 raw conversational sentences** (`8,000` samples per dialect) spanning **Beirut**, **Cairo**, **Doha**, **MSA**, and **Tunis**.

3.2 Data Hygiene & Exact Deduplication

Prior to splitting, rigorous data hygiene checks were conducted in `01_Data_Preparation.ipynb`:

- **Missing Values:** Zero null or empty records were found.
- **Exact & Post-Normalization Deduplication:** Removing duplicate records reduced the dataset from `40,000` raw samples to **38,963 unique cleaned sentences**. Removing duplicates *before* data splitting is critical to prevent identical sentences from appearing in both training and evaluation sets (**Zero Data Leakage**).

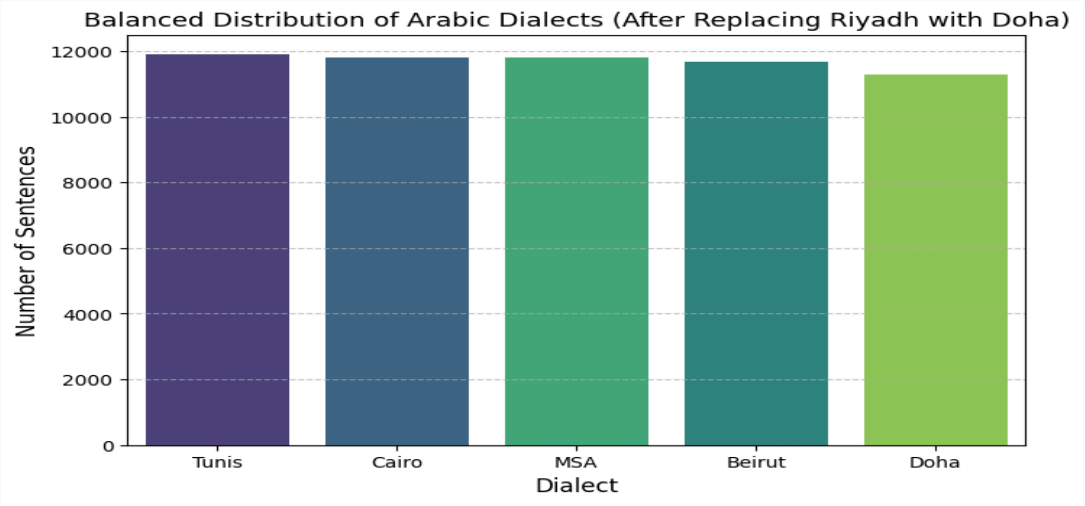


Figure 1: *Balanced class distribution across the 5 target dialects post-cleaning.*

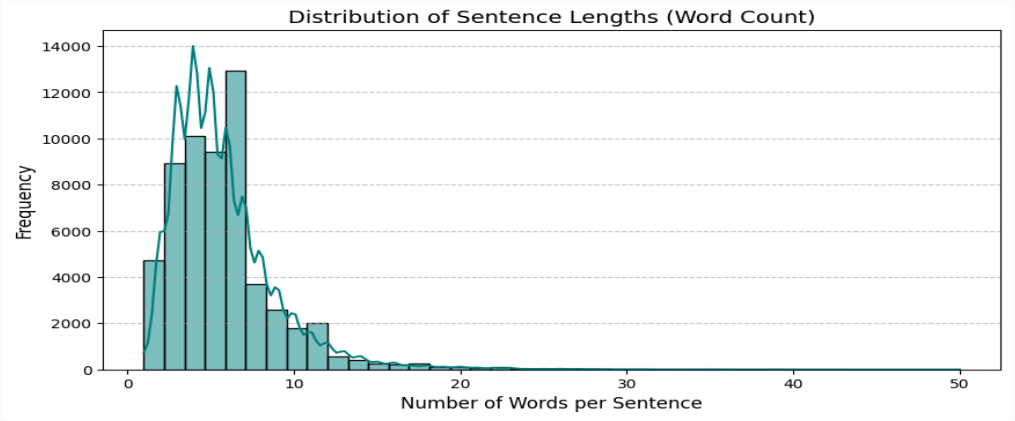


Figure 2: *Distribution of sentence word lengths showing concentration on ultra-short utterances.*

3.3 Key Linguistic Design Decisions during Preprocessing

Unlike conventional topic classification pipelines, two domain-specific rules were strictly enforced:

1. **Retaining Arabic Stopwords:** Standard Arabic stopword lists strip functional words such as pronouns, negations, and interrogatives. In dialect identification, these exact words represent the strongest discriminative features (e.g., عايز / مش / إيه in Cairo vs. بدي / ما / شو in Beirut vs. برشا / شنوة in Tunis vs. ابغى / وش in Doha). Therefore, **no stopwords were removed**.
2. **Retaining Short Sentences:** Single-word and two-word expressions often carry pure dialectal markers (e.g., معلش ,شورايك ,يعيشك). All valid non-empty sentences were preserved.

3.4 Orthographic Normalization Pipeline

A deterministic Regular Expression pipeline (`clean_arabic_text`) was applied consistently across training and deployment:

- Removal of Arabic diacritics (*Tashkeel*: `[\u0617-\u061A\u064B-\u0652]`) and kashida elongation (*Tatweel*: `\u0640`).
- Removal of URLs, digits, punctuation marks, emojis, and non-Arabic characters (`[^\u0600-\u06FF\s]`).
- Orthographic unification: ي ➔ ى, ا ➔ آ أ إ, and ه ➔ ة.
- Whitespace collapsing and trimming.

3.5 Stratified 70 / 15 / 15 Data Splitting

In `02_EDA_and_Data_Splitting.ipynb`, the `38,963` unique cleaned records were partitioned using stratified sampling (`random_state=42`) to preserve exact class proportions:

- **Training Set (`train.csv`):** `27,274` sentences (`70.0%`)
- **Validation Set (`val.csv`):** `5,844` sentences (`15.0%`)
- **Unseen Test Set (`test.csv`):** `5,845` sentences (`15.0%`)

4. Methodology & Model Architectures

4.1 Phase 1: Classical Machine Learning & Hybrid TF-IDF (`03_Baseline_Models.ipynb`)

To build a competitive non-neural baseline, we engineered a **Hybrid Feature Union** combining two TF-IDF vectorizers fitted strictly on `train.csv`:

- **Word N-Grams** `(1, 3)` **(`max_features=75,000`, `sublinear_tf=True`):** Captures complete dialectal words and multi-word idioms.
- **Character Word-Boundary N-Grams** `char_wb (2, 5)` **(`max_features=75,000`, `sublinear_tf=True`):** Captures morphological prefixes, suffixes, and subword roots inside agglutinated dialectal words.

Four individual linear classifiers (`Multinomial Naive Bayes`, `Logistic Regression`, `LinearSVC`, and `SGDClassifier`) were evaluated, followed by a **StackingClassifier Ensemble** (`cv=5`, `LogisticRegression` meta-learner) that combines the out-of-fold probability/decision outputs of all four base estimators.

4.2 Phase 2: Pre-Trained Transformer Fine-Tuning (`AraBERTv02` vs. `MARBERTv2`)

While TF-IDF relies on static bag-of-n-grams frequencies, Transformer encoders utilize **Multi-Head Self-Attention** to compute dynamic contextual representations of every token relative to the entire sentence. We fine-tuned two distinct Arabic BERT architectures:

1. **AraBERTv02 (`05_Arabert_dialect_final_model.ipynb`):**
   - Based on `aubmindlab/bert-base-arabertv02` (`136M` parameters), pre-trained primarily on formal MSA news articles and encyclopedic Arabic corpora.
   - Fine-tuned for `4` epochs with `learning_rate=3e-5`, `batch_size=32`, and `weight_decay=0.01`.
2. **MARBERTv2 (`04_MARBERT_FineTuning_and_Analysis.ipynb`):**

- Based on `UBC-NLP/MARBERTv2` (`163M` parameters), specifically pre-trained on **1 Billion dialectal Arabic tweets** alongside MSA data, featuring a dialect-rich WordPiece vocabulary (`100,000` subwords).

> **Anti-Overfitting Regularization Strategy:** During initial experimentation on `AraBERTv02`, validation loss began rising after Epoch 2 (`0.264` ➔ `0.305` ➔ `0.371`) while training loss dropped to `0.072`, indicating mild overfitting. To prevent this in `MARBERTv2`, we implemented a strict regularization protocol:

- Reduced peak learning rate to `2e-5` with a `10%` linear warmup ratio.
- Increased `weight_decay` to `0.05` and applied **Label Smoothing (`0.05`)** to prevent overconfident logit spikes on ambiguous short phrases.
- Activated `EarlyStoppingCallback(patience=2)` monitoring `eval_f1_macro` with `load_best_model_at_end=True`.

5. Experimental Results & Quantitative Comparison

5.1 Overall Progression Across All Models

Table 1 summarizes the progressive accuracy gains achieved across our experimental phases:

Phase	Model Name	Val Acc (%)	Val F1 (%)	Test Acc (%)	Test F1 (%)
Baseline 1	Logistic Regression *(Hybrid TF-IDF)*	89.37%	89.38%	89.70%	89.70%
Baseline 2	SGD Classifier *(Modified Huber)*	89.90%	89.90%	90.20%	90.20%
Baseline 3	Multinomial Naive Bayes (α = 0.3)	90.35%	90.33%	90.47%	90.45%
Baseline 4	LinearSVC (C = 0.5)	90.59%	90.59%	90.52%	90.52%
Ensemble	**Classical Stacking Classifier**	90.85%	90.84%	90.93%	90.92%
Deep Learning 1	AraBERTv02 Fine-Tuned	91.03%	91.00%	–	–
Deep Learning 2	**🏆 MARBERTv2 Fine-Tuned (Champion)**	**92.02%**	**92.01%**	**92.25%**	**92.24%**

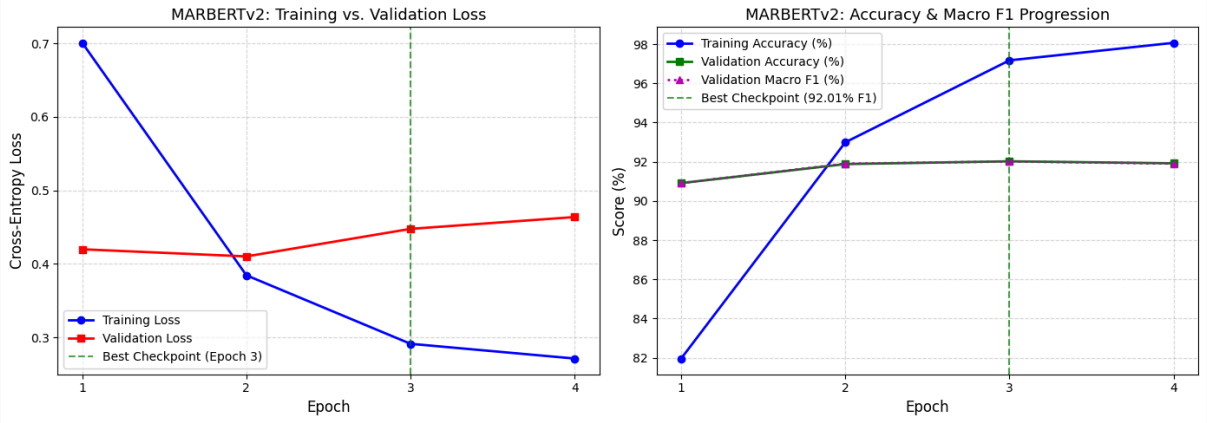


Figure 3: *MARBERTv2 fine-tuning loss curves and accuracy progression over 4 training epochs.*

5.2 Per-Dialect Breakdown of Champion Model (`MARBERTv2` on Unseen Test Set)

Evaluating `MARBERTv2` on the `5,845` held-out test sentences yielded highly balanced performance across all five dialects:

Dialect Class	Precision (%)	Recall (%)	F1-Score (%)	Test Support
Tunis (Maghrebi) 🇹🇳	95.21%	93.69%	94.44%	1,173
MSA (Modern Standard) 📖	93.60%	94.61%	94.10%	1,168
Cairo (Egyptian) 🇪🇬	92.42%	91.47%	91.94%	1,172
Doha (Gulf) 🇶🇦	90.02%	91.47%	90.74%	1,172
Beirut (Levantine) 🇱🇧	90.09%	89.99%	90.04%	1,160
Macro Average / Total	**92.27%**	**92.24%**	**92.24%**	**5,845**

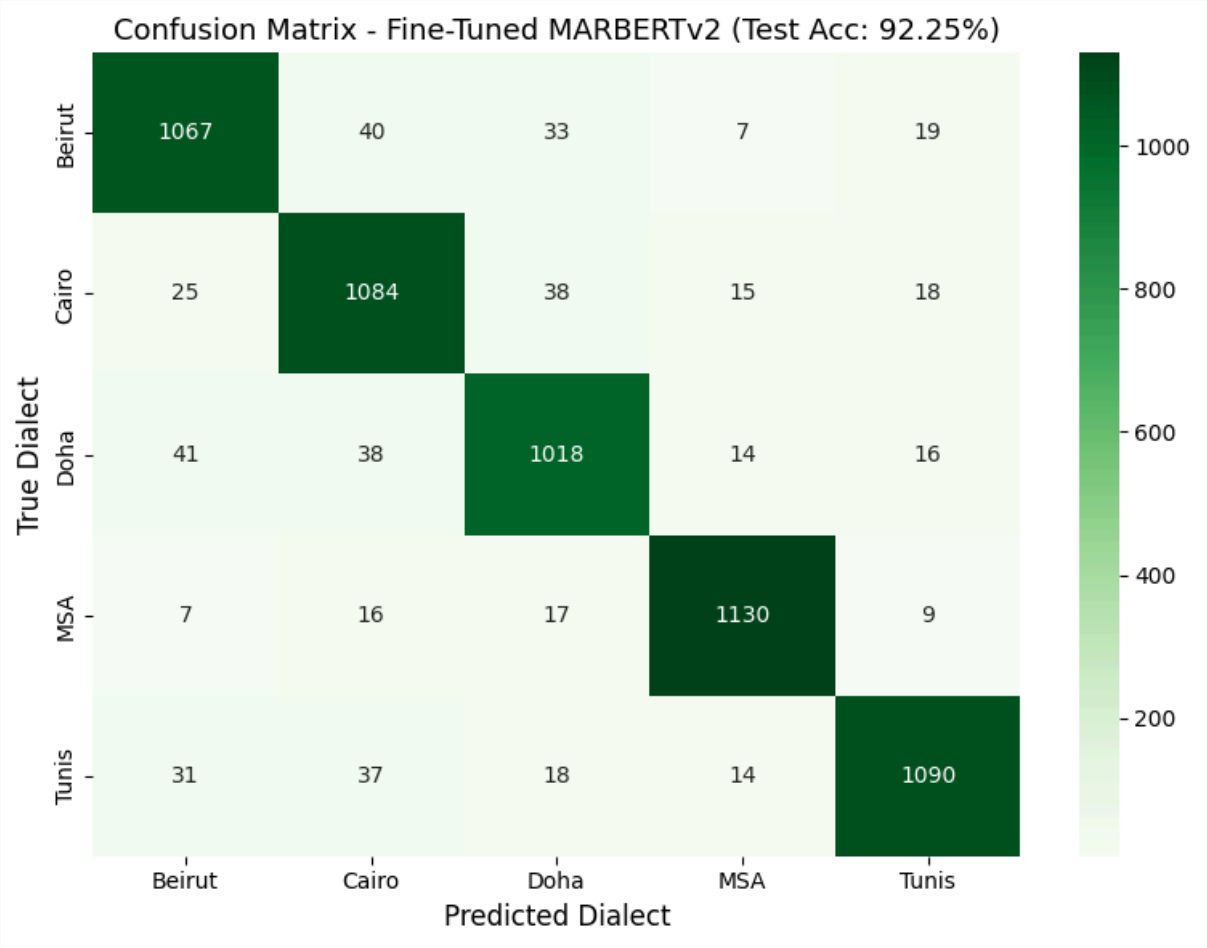


Figure 4: Normalized confusion matrix heatmap for fine-tuned MARBERTv2 on the unseen test set.

5.3 Why `MARBERTv2` Outperformed Both `AraBERTv02` and `Stacking`

- **Reduction of Total Errors:** On the exact same `5,845` test set, `MARBERTv2` reduced total misclassifications from `530` errors (in Stacking) down to `453` **errors**—correcting `77` complex sentences (`~14.5%` error reduction).
- **Subword Tokenization Advantage:** Because `MARBERTv2` was pre-trained on 1B dialectal tweets, its WordPiece tokenizer segments dialectal prefixes and suffixes cleanly without fragmenting slang words into meaningless character tokens, whereas `AraBERTv02` is biased toward formal MSA morphology.

6. Deep Error Analysis & Theoretical Dataset Ceiling

To understand why `453` sentences (`7.75%` of the test set) remained misclassified by `MARBERTv2`, we conducted an automated statistical and linguistic inspection in `04_MARBERT_FineTuning_and_Analysis.ipynb`:

6.1 Top Confusion Pairs

1. `Beirut` ➔ `Doha` (`48` **errors) &** `Doha` ➔ `Beirut` (`43` **errors):** Levantine and Gulf dialects share substantial conversational vocabulary when written without vowel diacritics.
2. `Cairo` ➔ `Doha` (`39` **errors) &** `Beirut` ➔ `Cairo` (`31` **errors):** Driven by pan-Arab conversational greetings and shared commercial terms.

6.2 Quantitative Proof of Context Deprivation (Sentence Length & Confidence)

Comparing the `5,392` correctly classified test sentences against the `453` misclassified sentences revealed two decisive metrics:

- **Average Word Count:** Correctly classified sentences averaged `7.21 words`, whereas misclassified sentences averaged only `3.86 words` (nearly half the length!).
- **Prediction Confidence:** Correct predictions achieved an average softmax confidence of `95.83%`, whereas misclassified predictions dropped to `77.57%` (uncertainty gap of `18.26%`).

6.3 Qualitative Inspection of Misclassified vs. Rescued Examples

- **Where** `MARBERTv2` **Rescued Classical Stacking Failure:**
  - *Sentence:* `"شورايك بعشره دولار للتلاته"` (True: `Beirut`) ➔ Stacking predicted `Doha`, whereas `MARBERTv2` **predicted** `Beirut` **(**`97.92%` **confidence)** by recognizing the agglutinated Levantine interrogative للتلاته + شورايك.
  - *Sentence:* `"انا ماشي فاصرف لي هالفيش لو سمحت"` (True: `Doha`) ➔ Stacking predicted `Cairo` (misled by `"انا ماشي"`), whereas `MARBERTv2` **predicted** `Doha` **(**`96.65%` **confidence)** via contextual understanding of `"هالفيش"`.
- **Why Remaining Errors Represent the Natural Dataset Ceiling:**
  - *Sentence:* `"تاني تقاطع عاليمين"` (Labeled `Beirut`, Predicted `Cairo`) — Identical phrasing used daily in both Cairo and Beirut.
  - *Sentence:* `"ايوه مسموح"` (Labeled `Doha`, Predicted `Cairo`) — Two-word affirmation common across Egyptian, Gulf, and Levantine speech.

> This proves that the remaining `~7.7%` error rate is not a model deficiency, but rather the **intrinsic linguistic overlap (Bayes Error Rate)** of ultra-short written Arabic phrases lacking audio phonetics.

7. Interactive Deployment (`Streamlit` Web Application)

To translate our research into a practical product, we developed and deployed an interactive web platform (`streamlit_app.py` & `app.py`) hosted via **Streamlit** and **Pyngrok** (`06_Streamlit_Interactive_App.ipynb`).

🌍 Smart Arabic Dialect Classification Platform

State-of-the-Art Transformer Models for Fine-Grained Arabic Dialect Identification (MARBERTv2 & AraBERTv02)

🎯 Single Sentence Analysis | ⚖️ Head-to-Head Comparison | 📁 Batch CSV Inference

✍️ Enter Arabic text to classify:

شورايك بعشره دولار للتلاته

🚀 Analyze & Predict Dialect

📊 Classification Results:

🎯 Predicted Dialect: **Beirut (Levantine)** 🇱🇧

📈 Confidence Score: **75.04%**

🧹 Normalized Text: شورايك بعشره دولار للتلاته

📊 Confidence Breakdown Across All 5 Dialects:

Beirut (Levantine) 🇱🇧 — `75.04%`

Cairo (Egyptian) 🇪🇬 — `21.45%`

Modern Standard Arabic (MSA) 📖 — `1.31%`

Doha (Gulf) 🇶🇦 — `1.12%`

Tunis (Maghrebi) 🇹🇳 — `1.08%`

Figure 5: Streamlit production application interface showing real-time single-sentence dialect analysis and confidence breakdown.

Key Application Capabilities:

1. **Automated Orthographic Normalization:** Raw user text is automatically cleaned using `clean_arabic_text` prior to tokenization.
2. **Single Sentence Analysis & Probability Breakdown (Tab 1):** Displays the predicted dialect, confidence score, normalized text, a smart linguistic alert when confidence `< 45%` (flagging dialect-neutral short phrases), and visual progress bars across all 5 dialects.
3. **Real-Time Head-to-Head Model Comparison (Tab 2):** Allows users and evaluators to test any sentence simultaneously on **MARBERTv2 (`92.25%`)** and **AraBERTv02 (`91.03%`)** side by side.
4. **Batch CSV Inference Engine (Tab 3):** Enables uploading a CSV dataset (`text` column), classifies all rows in batch mode on GPU, and provides a one-click UTF-8 CSV download (`dialect_analysis_results.csv`).

8. Conclusion & Future Work

This project successfully demonstrated an end-to-end engineering workflow for fine-grained Arabic Dialect Classification. By progressing systematically from a leak-free hybrid TF-IDF Stacking Ensemble (`90.93%`) to `AraBERTv02` (`91.03%`) and finally a regularized `MARBERTv2` **(`92.25%` Test Accuracy)**, we established that domain-specific pre-training on dialectal social media text provides a decisive advantage in resolving morphological agglutination and contextual ambiguity.

Future Directions:

1. Expanding classification coverage from 5 city/regional dialects to 25+ fine-grained Arab cities (MADAR-26).
2. Integrating acoustic/speech features (Multimodal Audio + Text) to resolve short phrases whose written form is identical across dialects but whose pronunciation differs.
3. Quantizing `MARBERTv2` using ONNX / INT8 quantization for ultra-low-latency edge deployment.